CSCI 441 - Software Engineering

SuperAsk.AI: AI-Powered Government Service Assistant

Team AO3

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Project URL: https://github.com/Monunhh/Super-ask.git

Team profile:

1. **Yean Fong**: project management, organizational skills, assertive communicator. Proficient in programming languages such as Python, C++, JavaScript, HTML, CSS, Design.
2. **Phallamonunhh Chun**: UX/UI, Business Analysis, Cybersecurity, CSS, C++, AI
3. **Bunkheang Heng**: Technical development, System architecture, Full Stack Development, Database Design, Machine Learning.
4. **Doungkhe Neang**: UX/UI, data curation and management, technical documentation, and information organization.
5. **Heejae Choi** : UX/UI, interface planning, programming languages such as C++, JavaScript, HTML, CSS, and documentation.
6. **SereyProphea Sarin** : Documentation, communication, UX/UI, Java, JavaScript, HTML, CSS

Roles

- **Yean Fong: Project Lead** – coordination, ministry liaison, timeline
- **Phallamonunhh Chun: AI/ML Engineer** – RAG pipeline, model integration, guardrails
- **Doungkhe Neang: Data/Content Curator** – sourcing, verifying, and tagging official content
- **Bunkheang Heng: Web/API Developer** – chat UI, web app, API, Database
- **SereyProphea Sarin: QA/Testing Lead** – accuracy testing, bilingual QA, user feedback collection
- **Heejae Choi: UX/UI Designer** – designing user flow, chat interface, page layout, and overall usability

1. Project Description

Introduction

SuperAsk is a generative AI chat service that answers Cambodian citizens' questions about government procedures in Khmer and English. Available as a standalone website, it gives free, registration-free, 24/7 answers on procedures, required documents, fees, timelines, deadlines, and office locations. This matters because access to accurate government information is a basic condition for citizens to exercise their rights and use public services efficiently, yet that access currently does not exist in a reliable, centralized form.

Problem Diagnosis

Government procedural information in Cambodia is scattered across 100+ institution websites, Facebook pages, Telegram channels, scanned PDFs, and counter notices; much of it in Khmer image format (not searchable) or outdated. As a result:

- Citizens make repeat trips, lose income, or miss legal deadlines (e.g., birth certificate registration) due to bad or missing information.
- Some procedures (justice system steps, sub-national services, social assistance) exist only as unwritten officer knowledge.

- Citizens turn to informal brokers who profit from the opacity.
- The government has no visibility into what citizens are confused about, and no way to prioritize fixes; the gap widens as more services go digital.

Proposed Solution

Intervention: An AI chatbot that answers *only* from content approved by designated ministry content stewards. Every answer cites the ministry, legal instrument, article, effective date, and last-verified date. If no approved source exists, SuperAsk declines and routes the citizen to a human officer rather than guessing.

A normal ChatGPT-style system is:

Question → AI generates answer

Our version should be:

Question → Identify intent → Retrieve approved government information → Generate answer → Attach source

Metrics for Evaluation:

- Answer accuracy rate (verified against official sources)
- Citation completeness (% of answers with full source attribution)
- "Decline-to-answer" rate for out-of-scope/unsupported queries (should be high when no source exists)
- Bilingual response quality (Khmer/English, including code-mixed input)
- User satisfaction / task completion in testing

Functional Requirements:

- Bilingual input/output (Khmer & English), including mixed-language queries
- Retrieval-augmented generation (RAG) restricted to an approved content database
- Mandatory citation display on every sourced answer
- Guardrails to keep scope limited to government procedures
- Fallback/escalation path to a human officer when no source is found
- Initial coverage: 4 ministries (Ministry of Public Works and Transportation, Ministry of Health, Ministry of Post and Telecommunications, Ministry of Interior)

Advantages and Benefits

SuperAsk.AI offers several advantages over the current way citizens search for government information. Instead of visiting multiple ministry websites, Facebook pages, Telegram channels, or government offices, users can access information from four ministries through one platform. The service is available in Khmer and English, does not require registration, and is available 24/7, making government information easier and faster to access.

Another important advantage is reliability. Every sourced answer provides a citation to the relevant ministry or official source, allowing users to verify the information. If SuperAsk cannot find an approved source, it will decline to answer and direct the user to a human officer rather than provide an uncertain response. This can also reduce repetitive inquiries handled by government staff, allowing officers to focus more on cases that require human assistance.

SuperAsk may also benefit government ministries by showing which procedures citizens frequently ask about or find confusing. Aggregated query data could help ministries identify areas where public information is unclear or outdated. In the longer term, the same model could be expanded from the initial four ministries to additional ministries and government services.

Potential Challenges

Despite these advantages, SuperAsk also has several potential challenges. The initial version only covers four ministries, meaning many government services will still remain outside the system. In addition, the accuracy of the system depends on ministries regularly reviewing and updating approved content. If policies, fees, documents, or procedures change and the information is not updated, users may receive outdated guidance.

Accessibility is another limitation. Citizens who have limited internet access, do not use smartphones, or are unfamiliar with digital services may have difficulty using SuperAsk. Khmer-language and mixed-language questions may also be more difficult to process accurately because fewer language-processing tools are available for Khmer compared with English.

Another challenge is determining when the system should answer and when it should decline. If the system is too cautious, users may not receive useful answers, while if it is too permissive, there is a greater risk of providing information that has not been properly verified. Therefore, SuperAsk will require continuous testing, content maintenance, and cooperation from ministry content stewards.

Overall, the benefits of SuperAsk can outweigh these limitations if the system is properly maintained, regularly updated, carefully tested, and designed to remain accessible to different groups of citizens.

Evaluation

SuperAsk will be evaluated in four main areas: answer accuracy, guardrail and fallback behavior, bilingual performance, and usability.

First, answer accuracy will be tested using realistic citizen questions. Each response will be manually compared with the official source cited by the system to determine whether the information is correct.

Second, the guardrail and fallback system will be tested using questions that are unsupported or outside the scope of the approved government information. The system will be considered successful when it correctly declines these questions and directs users to a human officer instead of guessing.

Third, the bilingual capability will be evaluated using Khmer, English, and mixed-language questions to determine whether SuperAsk can understand the user's request and provide an appropriate response.

Finally, usability will be evaluated through a small pilot group completing realistic tasks and providing feedback on whether information was easy to find, understandable, and trustworthy. Testing will take place throughout development, including before the first demo and again across all four ministries before the second demo.

2. Plan of Work and Product Ownership

Plan of Work

Phase	Activities
1. Scoping	Confirm ministries, gather approved source documents
2. Data Collection	Build web crawler + manual content curation pipeline
3. Content Vetting	Ministry content stewards approve/tag source material
4. Build	Develop RAG pipeline, guardrails, citation system, and bilingual handling
5. Testing	Internal testing, friends/family user testing, accuracy checks
6. Iteration	Refine based on feedback, expand ministry coverage

Tools

- **Programming/Frameworks:** Python, LLM API (e.g., Anthropic Claude API), RAG framework (e.g., LangChain/LlamaIndex), web framework for frontend (React)
- **Data Resources:** Web crawlers of official ministry sites/Facebook pages; manually curated/verified legal documents (Prakas, circulars)
- **APIs:** LLM provider API, translation/language-detection tools for Khmer-English handling
- **Infrastructure:** Database for approved content + citations, hosting for web app

Measure of Success

- The AI answers test queries accurately and only from approved sources (verified manually against official documents)
- Smooth Khmer/English language switching and mixed-language input handling
- No fabricated answers - SuperAsk correctly declines when no approved source exists
- Positive feedback from internal testers and a small pilot group (friends, family, sample users)

Weekly Schedule

Course Week	Project Work	Project Deadline
Before Week 4	Finalize project idea, team profile, proposed solution, functional features, and plan of work	**Project Proposal- Sep 10, 2026**
Week 4	Complete statement of work (SOW) and requirements. Complete functional requirements specification and UI	**Report #1 - Part 1 & 2: Sep 21- 25, 2026**
Week 5	Finalize requirements, SOW, and Report #1	**Report #1 - Full: Sep 30, 2026**
Week 6	Develop interaction diagrams; continue data collection/content vetting	**Report #2 - Part 1: Oct 6, 2026**
Week 7	Develop class diagram and system architecture; continue UI/RAG development	**Report #2 - Part 2: Oct 12, 2026**
Week 8	Integrate UI, bilingual interface, RAG pipeline, guardrails, and documentation	**Report #2 - Full: Oct 21, 2026**
Week 9	First demo implementation and integration testing	-
Week 10	First demo testing, debugging, and preparation	-
Week 11	Present working system and collect feedback	**Demo #1: Nov 9, 2026**

Week 12	Full integration, testing, and prepare Report #3	**Report #3 - Part 1: Nov 16, 2026**
Week 13	Complete system testing and final Report #3 documentation	**Report #3 - Full: Nov 23, 2026**
Week 14	Refine system based on feedback; prepare final demo and reflective essay	-
Week 15	Final system demonstration and finish reflective essay	**Demo #2 - Dec 4, 2026**